What If, We Live Forever Digitally?

Investigating the Future of AI-based Digital Afterlife. A Speculative Design Approach.

AASHAY A SHIRSAWADE

Department of Informatics and Media, Uppsala University, Sweden, aashay-ajit.shirsawade.2940@student.uu.se

The thesis delves into the concept of digital afterlife and its potential impact on the relationship between the deceased and the bereaved. Here the study explores the notion of being digitally immortal to stay in touch with the people left behind. It uses speculative design to envision future technologies that could enable digital resurrection, allowing individuals to interact with virtual representations of their loved ones. A participatory design workshop using speculation as a thought experiment was organized to create artefacts using a generative Artificial Intelligence (AI) program. These ideas and digital artefacts were set in the preferred future using the "what if" narrative to discard the aspect of feasibility and help participants visualize contexts and scenarios as well as open critical discussions. The study reveals various forms of digital afterlife, from realistic avatars in VR environments to abstract representations, and the importance of presence and immersion in creating a sense of connection with the deceased. Additionally, it discusses the tension between control and autonomy, as bereaved individuals seek to maintain power over the digital presence of loved ones, while also empathizing with the rights and autonomy of the deceased within the virtual realm. The speculative design approach facilitates critical discussions on the potential societal impact of digital afterlife technologies, highlighting the importance of responsible design and regulation. Finally, in accordance with the posthumanism lens, the complexity observed by the entanglement of the digital and the virtual realm has been demonstrated.

Additional Keywords and Phrases: Digital Afterlife, Speculative Design, Thanatechnology, Posthumanism, Continuing Bonds Theory.

1 INTRODUCTION

The concept of the digital afterlife, which involves AI-powered systems creating a virtual presence posthumously, is expanding rapidly [26]. Despite the complexities, there are hints that certain aspects of an individual's persona could be encapsulated in a program, allowing interaction with future generations [4]. Predictions by experts like Ray Kurzweil and Hans Morovec suggest that advancements in artificial intelligence could lead to increasingly realistic avatars, potentially blurring the line between virtual and physical realms by the year 2040 [4].

In the past, much research has focused on legacy-making and remembrance through digital means [3, 17, 30]. These digital means encompass digital platforms and social media features engaging in death-related concepts of legacy-making [15], online obituaries [3], digital inheritance management [15], memorialization [3, 17], chatbots [26] and mind uploading [23]. Gulotta et al., reviewed 75 web-related systems designed to support practices regarding death which concerned remembrance, legacy making and bereavement [15]. Most of the systems (39) were designed to help support end-of-life planning E.g. last wishes. Another group of websites (22) were to aid the people who were left behind E.g. Obituaries. The third category of websites centred around genealogical research and relationships where these websites help understand one's ancestorial legacy to figure out their self and legacy [15]. Furthermore, the creation of digital legacies, such as online memorials and digital memory boxes, dealt as an agency to preserve and share their memories, ensuring that the deceased are remembered and honoured. These digital legacies offer a form of digital immortality, providing a lasting tribute that can be accessed and interacted with by future generations [6].

Author's address: Aashay Shirsawade, Department for Informatics and Media, Uppsala University, Uppsala, Sweden.

This work was submitted in partial fulfilment for a master's degree in Human – Computer Interaction at Uppsala University, Sweden, on 12 June 2024. Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author must be honoured.

© 2024 Copyright is held by the owner/author(s).

The question remains, that with the advent progress of technology, as we are trying to achieve the afterlife through digital means, what will happen if the technology comes to reality? How could this technology be used, what forms it could take and how will it affect our current state? The research presented here aims to scope design in the field of Digital Afterlife with a particular nod to digital resurrection [3] to see its impact on how it is perceived by the people.

This thesis uses Speculative design to explore future implications of technologies by reflecting on present activities hypothetically [9]. The scope and ethics needed for the thesis make the speculative approach central to the thesis. The study is focused on the perceivers by speculating on the imaginative futures of how humans see themselves and other loved ones digitally resurrected. Considering the scope of design, the study will not conduct experiments on the available technologies but will be focused on creating artefacts of alternate futures and their consequences. By applying the theory of Continuing bonds [3, 17, 27] which states that the attachment between people who are left behind and the deceased does not necessarily have to be cut off but can be managed by helping the relationship to change its nature and evolve, thus, helping the bereaved move on in a healthier way. Thus, the study utilizes the theory of Continuing Bonds to understand how people would maintain a relationship with the deceased as bereavement is coped through a conversation [17]. Lastly, the theory of posthumanism was considered to interpret the dualism of living/dead and the existence of humans in the postmodern era to reconfigure and expand human possibilities [5].

1.1 Question

Based on past research and approaches, the "Digital Afterlife" phenomenon encompasses various aspects of remembrance, bereavement, and legacy-making. While these have been studied extensively by others recently, the question remains: what if advancements in AI technology remove the barriers to digitally resurrecting individuals, thus making it feasible? When it comes to thinking of the future where these technologies might take place affecting the functions and cultural meaning of being a part of it, the underlying question that motivates the study is: How would the advancements in Artificial Intelligence impact the relationship between the dead and the bereaved?

The thesis aimed to approach the study by Speculation in a participatory setting, thus uncovering how participants see themselves through different forms of digital resurrections and exploring its consequences.

1.2 Sub-Questions

1. How do advancements in AI and digital technologies influence people to envision themselves and others as digital immortals?
2. How does the concept of posthumanism challenge the dualism between the deceased and the bereaved?

1.3 Ethics

The topic was sensitive as it revolves towards exploring posthumous existence [17]. The discussion of death was crucial, and it was also important to consider the participant's personal experiences with grieving and mourning to gain a better understanding. The project's methodology involved creating fictional narratives by asking critical questions, but these narratives were designed to stay within the realm of fiction. This approach aimed to help people

set aside their personal feelings by using a medium that allowed for general opinions, ideas, and critique. Participants had the autonomy to sign up for the project where the invitation outlined the agenda and expectations. They were also free to withdraw from the study at any time. The data gathering followed GDPR guidelines (Appendix 11.1) to capture data like audio, pictures and generated artefacts.

2 BACKGROUND & THEORY

2.1 Legacy-making and Remembrance

Recent advancements in artificial intelligence and digital archiving have introduced new paradigms for posthumous presence[3]. Companies like Eternime [10] and LifeNaut [20] offer services that create digital avatars and memory archives from an individual's digital footprint, such as tweets, messages, vlogs, and photos. The practice of curating digital information to shape posthumous memories is increasingly facilitated by major digital platforms. For example, Facebook offers features like the Legacy Contact [15], allowing users to designate someone to manage their account posthumously, converting it into a memorial page. This functionality underscores the importance of organized digital artefacts—photos, social media posts, and other digital materials—in crafting a coherent and lasting legacy.

These digital avatars, sometimes referred to as forms of “cyberconsciousness”, “mind clones”, “mindfiles” or “mindware”, aim to provide likeliness of the deceased's personality that can interact with surviving loved ones, offering advice and maintaining a virtual presence [3]. Bassett [3] suggests that these digital personas can continue to learn and evolve, effectively taking on a life of their own. Because of the evident adaption of digital systems in everyday routines, the offline ways are digitally materialized creating online obituaries, using social networking sites for bereavement support and journaling illness experience thus increasing reminiscing and bereavement aid [15].

Despite the opportunities presented by digital legacy systems, there are significant challenges. With unequal access of people to their reach and understanding towards these technologies potentially creates disparities in how individuals can memorialize and be remembered [3]. Furthermore, many people do not engage with available digital systems for legacy-making, due to a lack of awareness, interest, or perceived relevance [3]. Researchers argue for inclusive design considerations that recognize diverse motivations and practices related to legacy-making [3]. These systems should support various ways people think about and engage with legacies, including those who may not feel a strong connection to traditional heritage or who wish to reflect privately on their lives [3].

However, when it comes to engaging with the posthumous heritage and the evolution of grief practices, it becomes evident that digital platforms like Facebook significantly enhance the communal and sociological aspects of mourning, as advocated by Kasket [17]. Traditionally, bereavement was often a solitary, inner experience, but social networking sites now allow for continuous, collective remembrance [17]. These platforms facilitate ongoing conversations about the deceased, enabling the construction of a shared, durable biography that reflects the deceased's life through the memories and interactions of their community [17]. This digital commemoration provides a sense of connection and comfort that surpasses traditional physical memorials, as participants find more satisfaction and vividness in virtual memorials [17]. Facebook's features, such as the ability to view all interactions

with the deceased and participate in post-death rituals, create a space where the deceased's social identity endures through communal engagement, thereby fulfilling the processes of grieving outlined in continuing bonds theory [17].

The intersection of digital technologies with remembrance and legacy-making presents both profound opportunities and substantial challenges. As digital systems continue to evolve, it is crucial to understand their impacts on the bereaved and society's perceptions of death and legacy.

2.2 Digital Afterlife technologies

Digital death has been an emerging area of interest in the field of HCI which [25] explores the topic of digital death and the role of technologies. The issue of "death studies" consisting of multiple clusters highlights strands observed in digital death [25]. Savin-Baden and Burden [26] separate these stakeholders into three sections: First, preservers, who make use of digital tech in legacy-making. This category includes not only living individuals but also individuals who may engage in preserving their digital legacy before their death. Second, receivers, are the ones who receive the legacies or digital traces in the form of artefacts or possessions left behind. Lastly, mediators, are those who work for the transitioning of such digital possessions by looking after the guidelines and beliefs in the technical and cultural contexts [25, 26].

In "Virtually Human" Rothblatt [23]explores the concept of "cyberconsciousness", where human consciousness can be replicated using software and computer technology. The book discusses Bina48, an advanced robot created by Hanson Robotics and a "Mindclone" of Rothblatt's wife, capable of engaging in complex conversations and expressing spontaneous thoughts based on a digital Mindfile that contains the real Bina's thoughts, memories, and emotions. Rothblatt [23] explains that by actively using social media platforms such as Twitter, Facebook, Instagram, or other social networking sites, individuals are essentially creating a mindfile—a digital backup of their mind. This Mindfile could potentially be brought to life using Mindware, software that mimics human brain functions, allowing digital entities to process information, experience emotions, and attain self-awareness. This technology, seemingly a part of science fiction, is already in development, with significant global efforts and substantial investment in the Brain Activity Map project [23].

Savin-Baden and Burden [26] explore the technological developments in the field of digital afterlife and its ramifications. The primary observations of the paper are the ways of utilizing digital technology for commemoration and memorialization, practices of grieving and mourning digitally and immortalization technologies. They described various techniques the digital technology is used e.g. Growth in veneration culture, bringing the physical and digital together creating a restless dead (living headstones, augmented coffins, urns embodying the head of the deceased), the rise of chatbot-based technologies, its capabilities and implications of learning and sustaining infinitely are argued by the author [26]. While most of the technologies are focused on the creation of immortals, management of the receivers and its ethicalities remain unexplored [26].

Pitsillides [21] focused on the bereaved and the role of the designer in creating an agency between them and the deceased where the emphasis on mixing digital and physical interactions for mourning practices can be seen [21, 25]. Within the same issue, she explores in depth "Designing with things" where the notion of expanding the power

of design creates new forms of agency between the deceased and the bereaved. The paper sheds light on different rationales of artefacts being digital, physical and hybrid which creates different ways for the bereaved to make meaning out of it, thus creating a new form of agency [21].

2.3 Theories

2.3.1 Continuing Bonds Theory

The concept of continuing bonds, as discussed by Kasket [17], underscores the importance of social and communal interactions in the grieving process [3, 16, 17]. Traditionally, these bonds were maintained through physical memorials and rituals, but the advent of digital technologies has transformed these practices. The emphasis is that bereavement is not solely an individual experience but is deeply rooted in social and cultural contexts [16, 30]. This aligns with the argument that constructing a durable biography of the deceased requires accurate and shared memories, facilitated through conversations with others who knew them. However, modern mobility can challenge these interactions, a challenge now mitigated by social networking sites. Empirical evidence from studies [16, 17] supports the effectiveness of digital platforms in aiding the bereavement process. Participants in research studies express a preference for visiting virtual memorials over physical cemeteries, finding them more satisfying and connected [22]. This preference highlights the unique comfort derived from the interactivity and accessibility of digital memorials. For instance, within the study of Klass, [19] a participant named Ava noted that posting on her deceased friend's Facebook wall provided a comforting sense of ongoing connection, exemplifying the communal continuing bonds that [19] emphasized. Supporting the studies, [30] argued that there is a distinction between aiding or hindering the adaptation of continuing bonds. The results showcased that while continuing bonds can be part of healthy grieving, it's not universally beneficial and can be particularly challenging for individuals who face sudden, unexpected losses [30]. The study calls for a nuanced understanding of continuing bonds in bereavement and tailored support for those who struggle with strong attachments to the deceased [19, 30].

Additionally, She et al., [27]explore bereavement from the perspective of frontstage and backstage grieving. The study incorporated the theory of continuing bonds where they were focused on a specific disorder called Prolonged Grief Disorder (PGD) [27]. This disorder is observed when the individual is experiencing grief for a stretched amount of time than usual thus needing support [27]. With the proliferation of social media and digital technology aiding bereavement on the front stage, what [27] observed was a challenge where a mourner has an isolated backstage grieving process which is also important to work for healthy grieving and thus helping the continued bond with the deceased to be positive [27]. The study employed methods of diary study and semi-structured in-depth interviews with participants engaging in "a custom-made web-based application to honour the deceased person's memory" called Living Memory Home (LMH) [27]. The results point out the scope for design opportunities for backstage grieving with limitations calling in the need for more comprehensive research that includes diverse cultural and religious contexts, a broader participant base, and extended study durations to better understand and design digital technologies for grieving [27].

2.3.2 Posthumanism

The study examines the relationship between the deceased and the living utilizing technological means that aligns with the principles of posthumanism [5]. Posthumanism defines an approach to understanding human interaction

with the natural environment [5, 12]. This approach provides an epistemological foundation that is not centered on humans and their core values thus critiquing the anthropocentric analysis of the subject at hand [5]. Bolter [5] argues that at its core posthumanism rejects Cartesian dualism and challenges traditional modernist principles to encourage a rethinking of our understanding of art, culture, and society [5]. This perspective opens various interpretations of innovation and tradition shifting the focus from the user-centred paradigm towards understanding complex socio-technical systems that could be a part of our human engagement [5, 12].

3 METHODOLOGY

3.1 Speculative Design

The thesis approach of Speculative Design is inspired by Dunne and Raby [9] - "Speculative Everything: Design, Fiction, and Social Dreaming" where they put a strong focus on the ideas of possible futures as tools for challenging and anchoring current state to emphasize the wants and dislikes from the future [9]. Approaches in the design industry of today's world are oriented towards industrial visions whereas Dunne and Raby [9] argue for a shift in design towards challenging the social structures, values, technological grounds, and cultural trends [9, 14]. This investigation's approach to Speculative design falls under the constructivist paradigm [1, 29]. The explorative approach has been chosen deliberately to inform the exploration of the above-mentioned research problem of exploring the possibilities of presence/resurrection in the AI-based digital afterlife. The speculative design approach helps a designer empathize with the public in general to understand their vision and expectations operating with technology cast in the future [31], which in the thesis context are technologies that help them resurrect as a digital beyond.

3.2 Participatory Speculative Design

The idea implies collaboration as speculation is often brought up to be practiced in a participatory environment [31]. While speculative design's complete adoption of a 'participatory turn' remains uncertain, a noticeable trend is emerging where many speculative designers are integrating participatory modalities and co-design techniques [8]. Motivations for involving non-designers vary, encompassing normative and pragmatic objectives [11]. Some aim to amplify marginalized voices, ensuring those affected by future outcomes towards a contribution [11]. Others challenge privileged perspectives, seeking diverse narratives [11]. Involving non-designers informs the process with a grounded quality, encouraging social debates on the future as a mutual learning experience [8, 11]. This synthesis forms a crucial chapter in the evolving discourse on participatory design within the speculative design landscape.

From an epistemological standpoint, the participatory turn involves the examination of knowledge that arises through interaction [29]. Thus, participatory engagement was utilized to create tangible artefacts by collaboratively thinking of a reality participants desire where the artefacts were used as probes to spark debate and discussions on its effect on global issues [18].

3.3 Thought Experiment

Thought experiments are defined as constructions crafted from ideas expressed through design, serving as tools to think about challenging issues and allowing the full utilization of imagination [1, 9]. The study also aligns with Thanatechnology, a term coined by Sofka [28], describing the engagement of death and technology [3] through a strong focus on the Speculative Design approach [8]. Within the context of this thesis, this has meant to initiate a workshop of thought experiments [1] with a multidisciplinary group of scholars in the fields of design, sociology, anthropology, psychology, and IT.

The imaginative aspect of thought experiments is central to Dunne and Raby's vision of speculative design [9]. The main reason to employ a cross-disciplinary team is that the setting will spark better ideas by providing an understanding of the niche to work within. Secondly, with a limitation of analyzing the data with the affecting observed audience of grievers, mourners and bereaved, speculative approaches consider “what-if” scenarios making the people empathize with the situation challenging the complex narratives towards very fundamental [9, 30].

Building on the work by Barendregt and Vaage [1], thought experiments are valuable tools in speculative design, aiding designers in conceptualizing and sharing their designs. They extend beyond math, science, and philosophy to include ethics and help reveal inconsistencies in theory and practice, leading to paradigm shifts [1]. The approach will try to align thought experiments in a participatory workshop setting as a tool to allow multiple conclusions to be debated upon rather than just stating a dichotomous assessment of good and bad or accept or reject.

3.4 Data Gathering Method

The data collection was performed by a co-design workshop (Figure 1). The workshop was to explore multiple trajectories where people imagine themselves as AI-generated digital afterlife artefacts. For the co-design workshop, cross-disciplinary teams were organized. Convenience sampling [2] was leveraged to gather participants who were invited to attend a the workshop (Appendix 11.2). 13 master’s students attended the workshop coming from backgrounds of Psychology (2 participants), Computer Science (1 participant), Anthropology (5 participants) and HCI (5 participants). The utilization of speculation within a participatory co-design workshop was anticipated to gain a more nuanced understanding thus resulting in the depth and span of the research findings. The practice revolves around generating artefacts of the future(s) by considering the technologies of today by fostering speculation to extract abstract imaginations for new perspectives on such problematic issues to be acknowledged [1, 18, 24]. The workshop was planned in the given format:

1. Introduction
2. Icebreakers.
3. Brief ppt on related topics.
4. Generating Proposals of the Future for Digital Immortality.
5. Creating artefacts with the help of AI generative platform "MidJourney".
6. Presentation of Artefacts.
7. Critical Discussion

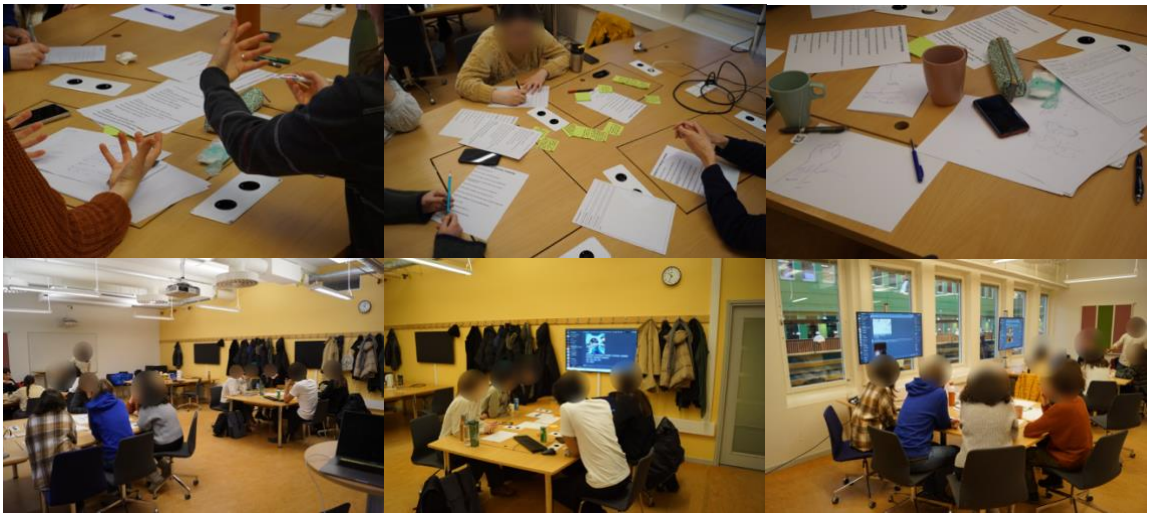
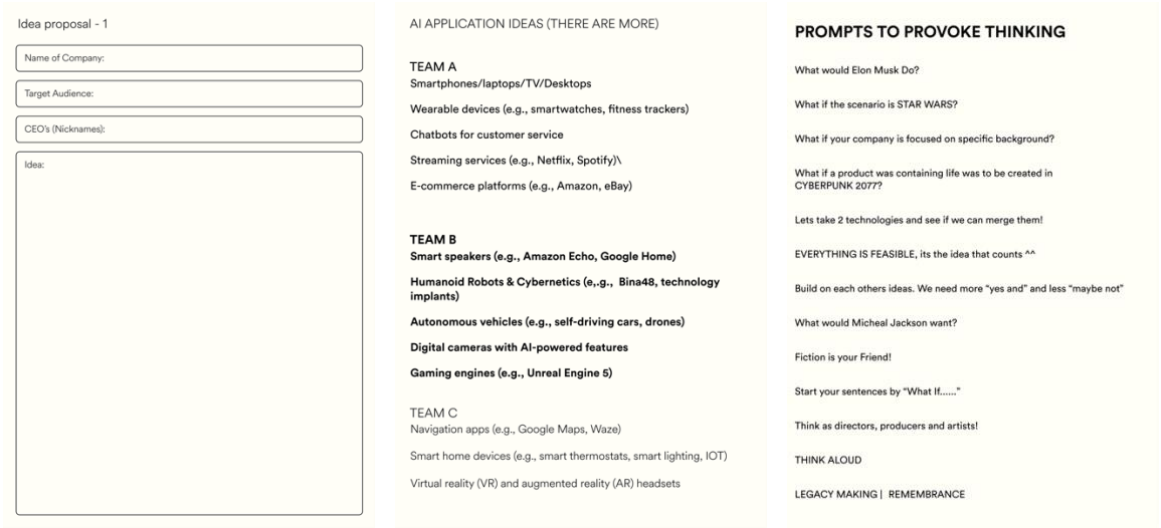


Figure 1: Participatory workshop engagement.

In making the contexts conventional, approachable and product-oriented, the thesis workshop opened with an introduction to various topics of AI, previously performed studies, and existing technology adding up to participants' understanding, thus, bringing them at shared understanding. As the workshop expected participants to think about artefacts or products of the future, envisioning preferable futures based on the existing technologies, the lived experience helped inspire creativity and acceptance towards new ideas giving a plethora of agency to create artefacts. Participants were randomly divided into 3 teams of A, B and C for a thought experiment. They were asked to imagine themselves as future CEOs creating products that could represent an Afterlife. The workshop began with participants generating several ideas about how an afterlife could look. They brainstormed different ways to encapsulate a digital afterlife. Later, they narrowed down these ideas to one or two concepts and provided context, deeper meaning, and background information to help explain their ideas to others in later stages. One attention was to not have overlapping ideas across the teams; thus, the teams were given value cards (Figures 2, 3 & 4) focusing on different technologies revolving around the Digital Afterlife such as AR, VR, AI, and handheld products [13]. Not to restrict them with the value cards, they were open to having inspiration from other team values as inspiration and for integration. In the case of the thesis, due to the limitation of material and skill needed to craft an artefact, an online AI text-to-image generative platform, Midjourney, was used to generate artefacts. These artefacts were created through several attempts to build on previous ideas through comprehensive discussion (E.g. Figure 5) for the rest of the idea progressions (Appendix 11.3). Later, these artefacts were used as central to their ideas for explaining and critically discussing them with the rest of the participants. The data was captured through audio recordings across teams and was transcribed to analyze it. The discussions were also tied to the artefacts as they were being made thus giving data as visual representation to analyze and interpret it.



Figures 2, 3 & 4: Value Cards demonstrating proposal document, inspiration lists and prompts to provoke thinking.



Figures 5: Discussion and changes in prompts resulting in overtime progression of artefacts' clarity.

3.5 Data Analysis

The study focused on examining the impact of the artefacts created during the co-design workshop and the feedback gathered during the critique session during the workshop. This involved coding the data and conducting a Thematic Analysis. The data, which included texts, audio recordings, and AI-generated images, was transcribed while maintaining anonymity, and then coded to identify patterns and themes in the observed critiques.

The commonly used six-step framework of thematic analysis [7] will be used, which is:
1. Understanding the data.
2. General Coding.
3. Observing themes.
4. Analyzing themes.
5. Naming themes according to repetitions and interpretations of data.
6. Reporting.

4 RESULTS AND ANALYSIS

This chapter presents the outcomes of the conducted speculative participatory design workshop involving 13 participants, aimed at exploring the envisioning of AI technologies for resurrecting humans as AI entities. The workshop unfolded in three phases: participants first familiarized themselves with existing AI technologies, then engaged in creative ideation using Midjourney, an AI text-to-image generator, and finally, presented and critically discussed their ideas with the whole group. The workshop aimed to foster interdisciplinary dialogue and critical reflection on the ethical, social, and technological dimensions of AI-driven resurrection technologies. The results were showcased through artefacts produced by each team, accompanied by detailed explanations, highlighting the conceptual understanding of their ideas. These artefacts serve as tangible representations of participants' thought processes and catalysts for further exploration and reflection on the implications of AI technologies for our future.

4.1 Ideas

The team A discussion revolved around the concept of creating ideas around representations that were not based on a human form entity, but others called "Me and My Ghost," which involves two main ideas. The aim of first idea was aimed towards Reliving Past Rituals. The idea focused on reliving past rituals or activities that the user did with the deceased (Figures 6 & 7). The aim of the second idea was towards creating new memories with the deceased. The idea was devised around a gaming environment where aspects of the game involved creating new memories with a digital avatar representing the deceased (Figure 8). Other ideas discussed shortly around the table were an AI smart speaker trained in the voice of the deceased person and the other one was around neuro implant. This prior idea was that the person would have their consciousness uploaded to a speaker where it will be able to have normal communication like the way it is with current smart speakers. The latter one was where the consciousness exists in the mind and would not have anything to do with its representation to the outside world and would remain only in the sight and feeling of the person experiencing itself. These ideas were brainstormed in the initial phase but were not taken into prototyping.

The discussion in team B revolved around the technologies of AR/VR where the digitally resurrected entity will be a clone in a known VR environment but can also be augmented into the real world, experiencing activities together. The discussions were to create a realistic VR environment or AR projections and the details of sensations were put forth to create a real sense of being around a person (Figure 9 & 10).

The participants in team C developed a product that primarily evolved from an idea called ‘companion car’ that projects holograms of deceased loved ones to provide company and conversation through the car ride (Figures 11 & 12) and showcase a person being artificially around. The team also brainstormed other ideas where they could introduce a digital entity into a smartwatch as well to allow people to bring the deceased as companions with them.



Figures 6 & 7: Spirit animal representation & abstract representation of a deceased loved one.

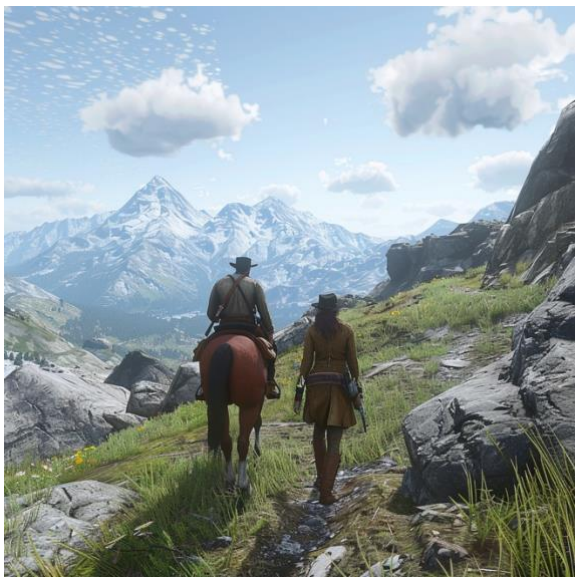


Figure 8: Red Dead Redemption inspired visual with a deceased friend on a quest.




Figures 9 & 10: Time-limited activity with a loved one in a specific environment.




Figures 11 & 12: Car AI idea representing the presence of the digitally resurrected hologram.

4.2 FORMS

Engagement within the teams had ideas that went into multiple direction encompassing different forms of resurrection. Even though the ideas differed, they consisted of three categorized similarities namely Representation, Presence and Immersion. The phrase 'form' defines the way of an artifact at the center of the scenario used to communicate the idea. The artefacts created in the workshop share the similar grounds on identified themes.

4.2.1 Representation

In the context of this thesis, "representation" points to the manner in which technology shapes the appearance and essence of a human within the intended afterlife. Throughout the workshop, participants explored diverse ways containing the presence of a deceased individual in digital form. These technological concepts were envisioned to withhold the essence of a human with an afterlife, taking on various formats as ideated by the participants.

The discussions predominantly revolved around representations that have a striking resemblance to humans. For instance, in the Car AI idea, there was a suggestion of projecting a hologram of the deceased individual in the passenger seat of a vehicle, encapsulating both their aesthetics nuances as illustrated by the quote: *"Maybe in the car, it can be like a hologram of a person."* Similarly, the VR/AR concept explored the possibility of interacting with a human projection in augmented or virtual reality environments, as depicted by the quote: *"so if you were in this VR environment [...] and then you could enter with the suit and you want to interact with your grandma."* These representations were crafted by sourcing data from various corners of images, videos, audios, and digital footprints to construct a clone or likeness of the individual being represented. Additionally, representations were not limited to strictly human forms. They also incorporated elements such as sound, as evidenced by the Speaker AI concept*: "Okay. So for example, Smart Speaker, what if you have like your partner who can speak to you like randomly? And then interact with you and have conversations with you?"*. Furthermore, representations extended to animal forms, as showcased by the Spirit Animal Hologram idea*: "something entirely different? Like, would it be like, something that doesn't look like that person? some kind of spirit animal live forever?".* Moreover, abstract entities were considered, evoking sensations of the person's presence, as indicated by the quote: *"So it's not, it's not their body that I'm working with, but it's somehow something they could have also been, like, some shape they could have taken doesn't necessarily need to be an animal".* The latter representations were conceptualized with a focus on encapsulating the essence or nature of the person. For instance, the Spirit animal representation was envisioned to be based on the deceased person's behaviour and nature, while abstract representations might incorporate elements such as the deceased's eye colour.

Observing the discussions, it became evident that participants either desired the human-like depictions to be extremely accurate or preferred a departure from strict realism, as they found the middle ground of resurrection to be unacceptable. This highlights the complexities and nuances involved in conceptualizing representations within digital afterlife contexts, which were further explored in the critique of motivations and implications underlying such choices.

4.2.2 Presence

While describing the technological aim to bridge the gap between the physical world and the digital representation of the deceased, the notion of presence was at the pinnacle. Ideas discussed contained subtle factors of presence, each emphasizing the deceased's essence and their continued presence in the lives of their loved ones.
Within the concept of a watch AI, it enables individuals to carry a digital memento of their loved ones wherever they go. By incorporating features such as voice or image projections, the watch cultivates a palpable sense of presence and engagement. The participant's reflection encapsulates this sentiment: *"I've also thought about like [...] always bringing your watch and maybe having some sort of moving video or something over of the person or that person's voice or anything. So I could always bring that person with me."* Here, the convergence of technology and personal mementos reinforces the feeling of having the deceased nearby, underscoring the significance of presence in the digital realm.

Similarly, in the context of the Car AI concept, the projection of the deceased within the vehicle with other features of a car withholds adds the experience of their continued presence. By intertwining the projection with speakers and imbuing it with the voice and characteristics of the deceased, the experience becomes more empathetic and lifelike. As articulated by a participant, *"so it felt like I was like driving around with that person next to me".* This integration of personal traits and memories into the AI persona further enhances the sense of presence, facilitating companionship and reminiscence during everyday activities.

Moreover, the exploration of haptic suits in VR/AR environments underscores the nature of presence. By incorporating tactile and olfactory sensations alongside visual and auditory stimuli, these technologies aim to create immersive experiences that evoke a profound sense of presence. The participant's consideration of utilizing a neuro- implant to simulate physical sensations, such as hugging, reflects this endeavor: *"what if I need something to feel. It could be like a wearable device. Some sort of an outfit that allows you to stimulate your muscles [...] hugging this person like these parts maybe feel yourself feel pressured heat in these areas."* Here, the integration of physiological responses enhances the authenticity of the interaction, heightening the feeling of closeness and connection with the deceased.

Participants contemplated the possibility of embodying the deceased within a virtual gaming environment, creating the character with either the characteristics or playing style of the individual. This notion extends the concept of presence beyond mere physical proximity, emphasizing the importance of evoking a sense of connection and familiarity with the deceased. The participant's reflection encapsulates this sentiment: *"What if you could like resurrect a person in like a game or as a game character? You could then also put some characteristics of the person."* Here, the idea of playing alongside a virtual representation of the deceased, whether through shared traits or familiar gameplay habits, underscores the immersive and interactive nature of digital afterlife experiences. It highlights the nuanced ways in which technology can evoke the feeling of having the deceased present, not just as a passive observer, but as an active participant in shared activities.

In essence, the incorporation of gaming elements into the concept of resurrection adds another dimension to the exploration of presence within digital afterlife technologies. It underscores the notion that presence is not solely contingent upon physical presence, but rather encompasses a broader spectrum of emotional and experiential

connections with the deceased. The integration of olfactory sensations serves as a common thread between the concepts of Car AI and haptic suits in VR/AR environments highlighting the multifaceted nature of presence in these technological contexts. Whether through personal mementos, AI projections, immersive sensory experiences, or virtual gaming interactions, the overarching aim remains consistent: to cultivate the digital realm with the essence of the deceased, fostering a sense of companionship and continuity for their loved ones.

4.2.3 Immersion

Immersion, as a sub-theme, encapsulates the niches of the specific mechanisms that gauge the sense of presence. While presence focused on the nature of the resurrection, immersion focused on the level of realism and its consequences contributing to the facilitation of the digital afterlife came up in the study.

One focal point of these discussions was the delicate balance between realism and detachment, emphasizing the need to create environments that are both immersive and distinguishable from tangible reality. For instance, participants highlighted the importance of crafting VR/AR environments that evoke a sense of presence while maintaining a distinct separation from the lived experiences of the deceased individual. As one participant noted, *"if you can create such a realistic group, you want to be able to separate yourself"* – a sentiment that underscores the necessity for immersion without compromising clarity of distinction from reality.

Moreover, discussions surrounding the representation of holographic projections underscored the complexity of balancing entity's presence from reality. Participants observed the challenge of creating representations that accurately reflect the essence of the deceased individual while ensuring that they remain identifiable as digital entities. This nuanced exploration of representation within immersive environments reflects a broader societal need for transparent delineation between virtual entities and tangible reality. As articulated by one participant, *"the reason we were trying to make them transparent is to show everyone that they were not real but holographic"* – highlighting the critical importance of maintaining a clear distinction between virtual entities and tangible reality.

Additionally, participants delved into the implications of realism in immersive environments, particularly as it pertains to the formation of memories. The discussion illuminated the nuanced interplay between realism and distortion, with participants grappling with the potential for hyper-realistic engagements to blur the lines between authentic experiences and digital simulations. As one participant pondered, *"how realistic is it to get? Because it's so easy to create false memories?"* – highlighting the inherent challenges in navigating immersive environments that closely mimic tangible reality. This exploration underscores the importance of fostering a critical awareness of the limitations and implications of immersion within digital afterlife technologies. Ultimately, the discourse surrounding immersion within the digital afterlife underscores the complex interplay between realism, representation, and memory formation, highlighting the need for nuanced considerations in the design and implementation of immersive environments.

Furthermore, considerations were raised regarding the potential consequences of hyper-realistic engagements, particularly in terms of memory formation. Participants recognized the importance of maintaining a pragmatic

mindset when navigating these immersive experiences, acknowledging the selective nature of engagements with digital representations and their potential impact on memory construction.

4.3 Temporal Engagement

The team discussions focused on the importance of time as a critical factor in their ideas and engagement. These ideas were seen in a preferable future setting and were critically discussed. As the aim was to see the direction of discussion while being critically discussed, time was an important factor that dripped in every team, while planning and in critical discussion. Time was seen through various formats as some described time as a context for the digital entity to resurrect and some saw time as a factor of being in the environment. The thesis recognizes these two formats of temporal engagement to be both structured and unstructured.

4.3.1 Structured Temporal Engagement

In the exploration of VR/AR afterlife and Car AI concepts, time emerged as a crucial element, containing the engagement with digital representations of deceased individuals within defined time boundaries. The VR/AR team discussed the notion of time as a dedicated period for interaction, where the sense of privilege and awareness can be seen by its finite nature: *"It's like a very privileged moment where you can exchange a bit and you are aware in your mind that you that you have 30 minutes for example."* The idea of being time-restricted interaction to not let the engagement have freedom of unlimited access as the issue of addiction was discussed and to make the VR experience separate from the real life. Even though the technology was stated to help cope with the loss of a loved one, the discussion around distortion of ones memory of the person and relying on the idealized version was put into the bracket of negatives of the technology. *"because if you create such a realistic environment, it can really affect the person itself, the real person [...] then we would have to limit people let's say once a month, or once a year, because I think otherwise, people might get too much into that environment".* This perspective underscored the importance of establishing limits within these technological frameworks to navigate their potentially immersive and addictive qualities.

The concept of time in these discussions served as a means of addressing the challenges inherent in engaging with digital afterlife technologies. Participants recognized the need to impose constraints on access to prevent overreliance and mitigate the risk of addiction, which could blur the distinction between authentic and fabricated memories. As one participant noted, *"we thought it would be time-limited, because we also don't want the person to be addicted to this feeling of being able to talk to their loved ones [...]".* This acknowledgement highlights the awareness of the therapeutic potential of these technologies, emphasizing the importance of fostering healthy coping mechanisms rather than facilitating escapism from reality.

Overall, the consideration of time in the context of VR/AR afterlife and Car AI concepts reflects a nuanced approach to designing digital experiences that balance immersion with responsible usage. By implementing time constraints, these ideas aim to provide meaningful opportunities for individuals to connect with deceased loved ones while promoting emotional well-being and resilience in the face of loss.

4.3.2 Spontaneous Temporal Engagement

In the exploration of digital afterlife experiences, on one hand, there is a recognition of the need to limit constant engagement with digital entities to mitigate the risk of addictive behaviors. As one participant reflected, *"Because certain habits might invoke feelings or memories. I think rather than being another volume of memory, it's more of a trigger to something that you reminisce."* This highlights a conscious effort to regulate the frequency of interactions, preserving the authenticity and significance of each encounter. Conversely, time also catalyzed evoking these digital entities during moments of emotional yearning for connection or companionship. This fluidity of time allows for dynamic interactions that feel more human and spontaneous, enriching the overall experience. As evidenced by discussions surrounding Spirit Animal AI and Car AI concepts, teams grappled with the question of when to summon the digital entity. Balancing the desire for constant presence with the need for natural spontaneity posed a significant challenge, leading to nuanced considerations of contextual triggers and personal rituals. This can be seen as a participant says: *"I want to say evoked by ritual and ritual can be Walking past something or making the coffee in a way that they used to and then it evokes that's what I want".*

The incorporation of music into the Car AI concept represents this approach to spontaneous temporal engagement. By integrating the deceased person's music choices and leveraging environmental cues, such as rain, the AI could suggest songs that evoke shared memories and emotions. As one participant suggested, *"With streaming services as you can have Spotify and you can Personalize them, […] they can actually be the one to suggest a song for you and you know that your mom or your dad, somebody would know it."* This personalized recommendation adds a layer of spontaneity and authenticity to the interaction, enhancing the overall sense of connection and reminiscence.

The concept of incorporating rituals into the experience further enhances its significance, evoking memories and emotions associated with shared moments or traditions. Whether it's walking past a meaningful location or engaging in familiar activities like making coffee, these rituals serve as anchors that deepen the connection with the digital avatar of your loved one. This intentional invocation of memories and emotions adds depth and meaning to the interaction, fostering a sense of closeness and continuity in the relationship.

4.4 Power Dynamics

During the discussion, participants raised concerns about control and autonomy. While the workshop initially focused on business proposals to create an afterlife, participants soon shifted their perspective to that of end-users while the proposals were critiqued. This raised concerns about the coexistence of virtually present entities within the physical realm. This section explains the outcomes of discussions on control and autonomy supporting the concept of the afterlife as a digital entity.

4.4.1 Control

While discussing the dynamics of control within digital afterlife technologies, the discussions showed the complexity between user agency and the presence of the digital entity. Participants engaged in conversations surrounding concepts of Car AI and Spirit Animal AI, where they expressed a desire for control over when and how these entities merge in their lives. In the context of Car AI, participants suggested regulating the entity's presence by increasing, decreasing, or completely turning it off to balance the need for privacy and autonomy with its

assistance. This can be observed when one of the participants says “So like a companion we are travelling. I guess you get bored while you drive. Yeah, exactly. Have to turn you off, now you're getting bored.”

Similarly, discussions on Spirit Animal AI highlighted the importance of managing the entity's appearance, ensuring it doesn't intrude unexpectedly. As one participant put it, *"We don't want them to appear out of nowhere [...] I'm always thinking like, if you would want to visit a place [...] and you don't necessarily want to take them there every time you go there"*. This reflects a broader concern for maintaining personal space and control over digital interactions.

Between the conversations surrounding turn-taking between humans and digital entities, the question of whether they should have human rights was touched upon. While privacy regulations were discussed emphasizing the need for space and control, there was a negative response to the idea after empathizing with digital entities. One participant even expressed, *"if that were to happen to me, and if my kids were switching me off, I would be sad."* This raises important ethical questions about the treatment and autonomy of these entities.

Finally, in the VR/AR idea, the fact that the deceased person cannot initiate contact further reinforces the notion of control, as the living person can decide when the interaction takes place. It can be seen by *“How do you as a deceased person have a right to be still person if your time is restricted, and you're not the one who can initiate the contact as the dead person?” who has the control? then how much are you willing to give that person also the authority to engage with you?”.* This statement captures the turn-taking between the living and the living dead. The control exerted by the person left behind created this power play where the rights are in the hands of the person left behind and not the digital entity. Next sub-section of autonomy focuses to determine how complex dynamics are seen from the point of view of a digital entity, as they become more prevalent in our society.

In essence, the discussions revealed the complexity of control within digital afterlife technologies, encompassing considerations of user autonomy, ethical responsibilities, and power dynamics. By delving into these deeper aspects, we gained insights into the unsteady nature of control and its implications for afterlife technologies.

4.4.2 Autonomy

In the discourse surrounding the agency of the digital entity, discussions emerged regarding its capacity for generativity and privacy. A central consideration was whether the AI should be afforded the autonomy to evolve and develop its own identity or remain strictly reflective of the person's characteristics prior to death. This notion extended beyond individual team ideas, encompassing broader considerations on the existence of these digital immortal entities.

Initial brainstorming sessions emphasized the thought that humanity entails openness to new experiences, suggesting that AI should possess the ability to think and grow. One participant articulated this viewpoint by stating, *"if you have enough data, you could emulate the effectiveness of our thinking. For example, personality, we have the openness to new experiences. So, you're probably more likely to want to experience new things and that could be integrated into the product."*. Later in the discussion, one participant made an interesting observation about the

difference between using a person's digital assets and their mind. They explained that using data to resurrect someone as an AI would only result in close guesses and that a person's true self is not just their digital footprint, but the sum of their life experiences. Therefore, the discussion on giving autonomy to an entity that lacks a real brain and is dependent on fragmented data raises questions about the reconsideration and tension between the rights and will to be given to the resurrected entity. The participant's statement was, *"I think there's a difference between using data to create it, no matter how much data you have it will still be used to 'guess what?' versus having a full brain upload which is uploading your consciousness.".* Thus, granting the digital entity autonomy to evolve raises profound questions about the future implications and understanding of these resurrected entities.

Furthermore, the conversation delved into the concept of defining a "death day" for the digital entity, acknowledging the significance of mortality in a human that attributes meaning to life: *"our mortality reminds us that we have this limited time to do things. "So, I'm also wondering like, how, how much of human nature can we include in something that is going to be immortal?"*. Participants pondered the implications of digital immortality, suggesting that a sense of mortality is essential despite its allure. This led to reflections on incorporating features that facilitate final goodbyes, such as a last car ride where individuals could bid farewell and interact with the digital entity one last time. This dialogue encapsulates the nuanced considerations surrounding digital entities' agency, identity, and mortality in the realm of afterlife technologies.

In tandem with discussions on control and privacy, considerations regarding the digital entity's rights were spoken for. Participants explored scenarios where the digital entity lacked autonomy to engage at its own will, prompting discussions on the necessity for designated spaces for their interaction or solitude. This discourse underscored the inherent tension between the rights and agency granted to these resurrected entities.

5 DISCUSSION

The thesis is focused on exploring speculative ideas as artifacts that serve as a catalyst for discussing our relationship to reality [9]. Although the technology to resurrect a human does not currently exist, significant strides have been made in the direction of a digital afterlife, as outlined in the chapters. The thesis delved into how individuals envision different forms of a digital entity and the implications for integration into reality, resulting in several interesting revelations as detailed in the results. The thesis utilized a participatory speculative design workshop where participants collectively imagined a preferred future and the technologies existing within that realm. Additionally, a critical discussion on the technologies included an empathetic exploration of the potential impact if they were to become a reality. While the background focused on the perspectives of current grievers and mourners and the depth of existing technology, the thesis aimed to explore the state of the future to challenge the notions of today and its potential effects on not only grievers and mourners but broader societal dynamics. Thus, addressing the research questions of *How do advancements in AI and digital technologies influence people to envision themselves and others as digital immortals?* and *how does the concept of posthumanism challenge the dualism between the deceased and the bereaved*? the following sections will provide discussions on the questions mentioned above with a reflection on the speculative design approach.

5.1 Complexity between being resurrected and experiencing resurrection.

The main body of work in this thesis focuses on exploring how people envision an afterlife, the form it takes, and the technology involved in demonstrating capabilities that prompt the mechanics and entanglement between the deceased and the ones left behind. The research showcases the complex interlacing of individuals who desire to be resurrected with the expectation that the experience will be real, contain feelings, emotions, autonomy to grow, and allow them to live on forever. On the other hand, when viewed from the perspective of those engaging with this digital entity, there is a desire to have more control over the resurrected personality. This complexity becomes evident in the context of immortality, revealing intricacies between the deceased and those left behind. It could be affirming where the simulated brief interactions with a deceased loved one could be a way to help those grieving say goodbye and cope with loss. However, the ethical considerations regarding this approach are complex pointing out the temporary comfort while engaging with an AI version of a person as it could potentially complicate the grieving process long-term.

This challenges the notion of continuing bonds where the theory speaks for changing nature and relation between the two to help a person grieve in a better way. Thus, bringing the deceased 'back' may prevent fully accepting their death. There were also uncertainties about how the AI system would develop and change independently over time being a positive thing as human nature is to evolve but being a computer-mediated simulation, the direction it can go cannot be judged where interacting with an altered version of the deceased years later could cause unforeseen distress.

5.2 Posthumanism

The study aimed to explore the interrelation between the physical and digital realms where the lingering digital consciousness can be observed as a non-human actor [12]. By conceptualizing the non-human actors engaging within the physical realm, posthumanism provided the foundation for the analysis of posthumous presence and its interplay, thus challenging the dualism of life and death [12]. The observed tension between control and power, along with the inquiries stemming from critical observations, shed light on the relational dynamics between human actors and digital entities as actants. The concept of the "second death," discussed within the workshop, underscores the profound ethical implications of requiring non-human components to demonstrate human-like commitments for posthumous existence. The discussions showed empathy towards virtual entities by not only prioritizing the needs of people but also acknowledging the rights of the deceased in a virtual realm. The discussion of rights for the deceased can be interpreted by the posthumanism lens that advocates equal footing challenging the notion of human superiority and centrality which is a classical anthropocentric approach [5].

The deceased's wishes may not align with the desires of those left behind to "keep them alive". This shows the complex emotional needs of the bereaved to have a loved one brought back as a digital presence but also stresses the importance of respecting the deceased's own identity and mortality if chosen not to be digitally immortal. Some individuals would still want to pursue life goals willingly knowing about immortality, rather than having an AI proxy fulfill an undefined existence. This also calls for the discussed sense of purpose to be alive. One of the participants noted that death reminds us of our limited human nature and how much would be inflicted if digitally an entity is immortal.

6 REFLECTION

Dunne and Raby, (2024) opened a perspective on speculative design that helped to explore the topic from a particular angle [9]. The artefacts created, dystopian as they sound and look, show how design can create narratives posing questions to new possibilities of what social, cultural and ethical implications of the technology can be. This aligns with the argument of utopias/dystopias by Dunne and Raby [9] where a dystopian world can be seen as a contrast to the real world that serves preventive narratives, alerting us to the potential future consequences if current trajectories are not critically examined and altered [9]. Participants openly discussed the once-whispered topic of death, sharing personal stories and critiquing emotions. The discussion raised concerns about the need for understanding and growth, drawing inspiration from marketplace dynamics and AI technology's advancements.

7 LIMITATIONS

Several limitations were identified that should be acknowledged. First, due to the sensitive nature of the topic, participants were selected with a focus on the diversity of their study domain to bring in different perspectives. Thus, the approach did not account for the diverse ideas and opinions that might arise from bereaved individuals or those who are terminally ill. Second, the study did not consider cultural and religious aspects, as it primarily focused on how people generally imagine and perceive the technology. Incorporating cultural and religious perspectives could have provided a different and possibly more nuanced view of the study. Lastly, the workshop engagement was limited to a short span, which could have been enhanced by conducting multiple workshops that built on the outcomes of previous sessions. This approach might have generated richer, more in-depth data and provided greater clarity to the ideas and artefacts produced during the workshops. Addressing these limitations in future research could lead to a more comprehensive understanding of the potential for digital technologies that aim to possibly provide digital immortality. Moreover, the work may also involve navigating relationships between the dead, the living, and hypothetical AI representations after death. This may involve the study of interlacing of the dead and alive, the judgement of realism or the governance of such technology as it is rapidly gaining traction.

8 CONCLUSION

The thesis offers crucial evidence of the ways people conceptualize a digital afterlife. In the virtual realm, as understood by the study, a person could live forever symbolically [3, 16, 26]. These afterlife representations can take various forms that capture the essence of the deceased. What's intriguing isn't just the representations themselves, but the implications that arise when the bereaved interact with the implied essence and how society perceives it. The entanglement of the posthuman entity within the physical realm draws various questions regarding who the technology should be designed for, how the power should be distributed and how the technology be must treated and regulated in its current state before nearing the thesis explore parallel futures. Even though the current practices illuminate crucial elements of memorialization, grieving, mourning and remembrance where these online practices are interconnectedly related to dying, what the thesis showcased is a parallel universe, an assumed possible and maybe even preferred future. Under the posthumanism understandings, the research revealed that the thought of being immortal digitally and engaging with a digitally immortal is complex. This demonstration of blurred lines between reality and virtuality, technology and beliefs, and dynamics of control, power and autonomy shows the complexity thus, expanding the future scope.

9 ACKNOWLEDGEMENTS

A special thank you to my supervisor, Sophie Landwehr Sydow, for helping me enormously at every step of the study. It was a pleasure working and having the support one could never ask for.

10 REFERENCES

[1] Barendregt, L. and Vaage, N.S. 2021. Speculative Design as Thought Experiment. *She Ji: The Journal of Design, Economics, and Innovation*. 7, 3 (Sep. 2021), 374–402. DOI:https://doi.org/10.1016/j.sheji.2021.06.001.

[2] Bark, I. et al. 2006. Use and Usefulness of HCI Methods: Results from an Exploratory Study among Nordic HCI Practitioners. *People and Computers XIX — The Bigger Picture*. T. McEwan et al., eds. Springer London. 201–217.

[3] Bassett, D.J. 2015. Who Wants to Live Forever? Living, Dying and Grieving in Our Digital Society. *Social Sciences*. 4, 4 (Dec. 2015), 1127–1139. DOI:https://doi.org/10.3390/socsci4041127.

[4] Bell, G. and Gray, J. 2001. Digital immortality. *Communications of the ACM*. 44, 3 (Mar. 2001), 28–31. DOI:https://doi.org/10.1145/365181.365182.

[5] Bolter, J.D. 2016. Posthumanism. *The International Encyclopedia of Communication Theory and Philosophy*. John Wiley & Sons, Ltd. 1–8.

[6] Burris, T. 2016. David E. Balk: Dealing with Dying, Death, and Grief During Adolescence: Routledge/Taylor & Francis Group, New York, NY, 2014, 265 pp, ISBN: 9780415534505. *Adolescent Research Review*. 1, 2 (Jun. 2016), 175–180. DOI:https://doi.org/10.1007/s40894-015-0002-z.

[7] Castleberry, A. and Nolen, A. 2018. Thematic analysis of qualitative research data: Is it as easy as it sounds? *Currents in Pharmacy Teaching and Learning*. 10, 6 (Jun. 2018), 807–815. DOI:https://doi.org/10.1016/j.cptl.2018.03.019.

[8] Clarke, R. et al. 2018. More-than-human urban futures: speculative participatory design to avoid ecocidal smart cities. *Proceedings of the 15th Participatory Design Conference: Short Papers, Situated Actions, Workshops and Tutorial - Volume 2* (New York, NY, USA, Aug. 2018), 1–4.

[9] Dunne, A. and Raby, F. 2024. *Speculative Everything, With a new preface by the authors: Design, Fiction, and Social Dreaming*. MIT Press.

[10] ETER9 Project: *https://www.eter9.com/*. Accessed: 2024-05-24.

[11] Farias, P.G. et al. 2022. Social dreaming together: A critical exploration of participatory speculative design. *Proceedings of the Participatory Design Conference 2022 - Volume 2* (Newcastle upon Tyne United Kingdom, Aug. 2022), 147–154.

[12] Forlano, L. 2017. Posthumanism and Design. *She Ji: The Journal of Design, Economics, and Innovation*. 3, 1 (Mar. 2017), 16–29. DOI:https://doi.org/10.1016/j.sheji.2017.08.001.

[13] Forlano, L. and Mathew, A. 2015. From Design Fiction to Design Friction: Speculative and Participatory Design of Values-Embedded Urban Technology. *Urban Informatics*. Routledge.

[14] Galloway, A. and Caudwell, C. 2018. Speculative design as research method. *Undesign*. G. Coombs et al., eds. Routledge. 85–96.

[15] Gulotta, R. et al. 2016. Engaging with Death Online: An Analysis of Systems that Support Legacy-Making, Bereavement, and Remembrance. *Proceedings of the 2016 ACM Conference on Designing Interactive Systems* (Brisbane QLD Australia, Jun. 2016), 736–748.

[16] Irwin, M.D. 2015. Mourning 2.0—Continuing Bonds Between the Living and the Dead on Facebook. *OMEGA - Journal of Death and Dying*. 72, 2 (Dec. 2015), 119–150. DOI:https://doi.org/10.1177/0030222815574830.

[17] Kasket, E. 2012. Continuing bonds in the age of social networking: Facebook as a modern-day medium. *Bereavement Care*. 31, 2 (Jul. 2012), 62–69. DOI:https://doi.org/10.1080/02682621.2012.710493.

[18] Kinch, S. et al. 2023. An exploratory study of using speculative artefacts in co-design. *CoDesign*. 19, 2 (Apr. 2023), 91–109. DOI:https://doi.org/10.1080/15710882.2021.2016847.

[19] Klass, D. 2006. Continuing Conversation about Continuing Bonds. *Death Studies*. 30, 9 (Oct. 2006), 843–858. DOI:https://doi.org/10.1080/07481180600886959.

[20] LifeNaut: *https://www.lifenaut.com/*. Accessed: 2024-05-24.

[21] Pitsillides, S. 2019. Digital legacy: Designing with things. *Death Studies*. 43, 7 (Aug. 2019), 426–434. DOI:https://doi.org/10.1080/07481187.2018.1541939.

[22] Roberts, P. 2004. The Living and the Dead: Community in the Virtual Cemetery. *OMEGA - Journal of Death and Dying*. 49, 1 (Aug. 2004), 57–76. DOI:https://doi.org/10.2190/D41T-YFNN-109K-WR4C.

[23] Rothblatt, M. 2012. THE TERASEM MIND UPLOADING EXPERIMENT. *International Journal of Machine Consciousness*. 04, 01 (Jun. 2012), 141–158. DOI:https://doi.org/10.1142/S1793843012400070.

[24] Rozendaal, M.C. et al. 2016. Exploring embodied speculation in participatory design and innovation. *Proceedings of the 14th Participatory Design Conference: Short Papers, Interactive Exhibitions, Workshops - Volume 2* (New York, NY, USA, Aug. 2016), 100–102.

[25] Sas, C. et al. 2019. Futures of digital death: Past, present and charting emerging research agenda. *Death Studies*. 43, 7 (Aug. 2019), 407–413. DOI:https://doi.org/10.1080/07481187.2019.1647643.

[26] Savin-Baden, M. and Burden, D. 2019. Digital Immortality and Virtual Humans. *Postdigital Science and Education*. 1, 1 (Apr. 2019), 87–103. DOI:https://doi.org/10.1007/s42438-018-0007-6.

[27] She, W.-J. et al. 2021. Living Memory Home: Understanding Continuing Bond in the Digital Age through Backstage Grieving. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama Japan, May 2021), 1–14.

[28] Sofka, C.J. 1997. SOCIAL SUPPORT "INTERNETWORKS," CASKETS FOR SALE, AND MORE: THANATOLOGY AND THE INFORMATION SUPERHIGHWAY. *Death Studies*. 21, 6 (Nov. 1997), 553–574. DOI:https://doi.org/10.1080/074811897201778.

[29] Spinuzzi, C. 2005. The Methodology of Participatory Design. *Technical Communication*. 52, 2 (May 2005), 163–174.

[30] Stroebe, M.S. et al. 2012. Continuing bonds in adjustment to bereavement: Impact of abrupt versus gradual separation. *Personal Relationships*. 19, 2 (Jun. 2012), 255–266. DOI:https://doi.org/10.1111/j.1475-6811.2011.01352.x.

[31] Tsekleves, E. et al. 2022. Exploring the use of speculative design as a participatory approach to more inclusive policy-identification and development in Malaysia. *Design Studies*. 81, (Jul. 2022), 101118. DOI:https://doi.org/10.1016/j.destud.2022.101118.

11 APPENDIX

11.1 Concent form

Please read this carefully.

There are no expected risks or consequences of participating in this project, other than the amount of time it might take to engage in the workshop. The project will collect and register information about you according to the EU's General Data Protection Regulation (GDPR) from Qualtrics and recordings by tools like Mobile Phones & Cameras.

The GDPR's first and second obligations (lawfulness, fairness, and transparency) ensure that all processes are transparent for participants, fair, and compliant with the law, and are explicitly explained.

In alignment with the third obligation stating that data should be collected and processed only for absolutely necessary research purposes and participants, some demographic information will be collected from you such as education level, and interests in subjective fields of technology. In addition, data will be collected about your discussions in the teams around the table. Lastly, the whole workshop will be recorded from table discussions to the artefact presentation and critique to have much detailed understanding.

Regarding the collected data and your privacy, in alignment with GDPR's obligations concerning data privacy, accuracy, and storage limitation, each piece of information will be anonymized using the nickname you will choose. Any identifiable details that might be collected will be removed such as blurring the faces and personal details shared across the discussion from videos, audios and images taken from the data to guarantee anonymization. The data will be securely stored on only the researcher's personal computer, limiting access exclusively to authorised researchers. Lastly, once the study is concluded, all data will be securely destroyed.

Your discussions and your results will be treated so that unauthorised persons cannot access them. Concerning the last GDPR's obligation, accountability stating data controllers are responsible for demonstrating GDPR compliance with all these principles, the person responsible for your personal data is Aashay Shirsawade.

According to the GDPR, you have the right to access your data in the project for free and, if necessary, correct any errors. You can also request that your data be deleted or that its processing be limited. However, the right to deletion and limited data processing does not apply when data is essential for the current research. If you want to access your data, contact the principal investigator by email. The investigator can be reached through email: aashay-ajit.shirsawade.2940@student.uu.se. If you're dissatisfied with how your personal data is processed, you have the right to complain to the research supervisor Sophie Sydow, email: sophie.landwehr.sydow@im.uu.se.

The purpose of the voice/video recordings is to make it easier for the researcher to extract information from what has been said. They do not serve another purpose. You have the right to refuse the storage of your voice recording without providing any reason. If you consent to the storage of the recording, you have the right to later withdraw that consent without providing any reason. Your voice recording will then be discarded. If you want to withdraw consent, contact the principal investigator at aashay-ajit.shirsawade.2940@student.uu.se

Your participation is voluntary, and you can choose to discontinue participation at any time. If you choose not to participate or wish to discontinue your participation, you don't need to explain why, and it won't affect your future care or treatment. If you wish to discontinue your participation, contact the person in charge of the project (see below).

The person responsible for the project is, Aashay Shirsawade.
Email: [aashay-ajit.shirsawade.2940@student.uu.se]

I, _____ have read the above and agree to participate in the study.

Nickname:

Signature:

11.2 Participant Gathering Invitation

Hello! My name is Aashay Shirsawade. A passionate student of the discipline Human Computer Interaction, Uppsala University overcoming complex problems with obvious solutions. This is an invitation/informative pre study for my thesis participant recruitment.

Please read this carefully. It will take around 5-7 minutes to finish. Thank you.

This project aims to investigate the possibilities of AI Digital Afterlife where immortality by resurrection can occur. We will be challenging this notion by a process called speculative design.

Digital Afterlife is a field that encompasses technologies like AI, IOT, VR and AR. In todays date, websites/companies like eternime.com and lifenaut.com claim to be building this technology for the near future. In a world of claims, no technology exists that makes a human immortal, neither physically nor digitally. The workshop's aim is to uncover those possibilities by designing artefacts of the future. WHAT IF these technologies make it possible for digital resurrection to occur? What would it look like? What are the forms it could take? What will we as human society do with it? What are its consequences on humanity and domains of industry overall?
This is where we as a team will engage in a co-design workshop where we will practise speculation to put ourselves in the near/distant/fictional futures to create artefacts and have a critical discussion around it.
There is no right or wrong in the workshop. It's all about perspectives and ideas that we bring to the table and design around it. When I say design, it means a lo-fi prototype of sketching, drawing, storytelling, AI image generation to add life to our ideas.
Here you will be making a grand entrance where you are asked to participate as you have a unique perspective coming from different study domain and a unique mind full of ideas. There were no specific criteria for participant selection rather being from your study domain.
The entire experiment will be conducted at a university building where you have to be physically present.

Workshop Details:

Date: 20th March

Time: [~08:30 - ~12:30] We will have breaks with Fika!

Location: Campus Ekonomikum, Room F224, Uppsala.

What's in it for you?

- Collaborate with experts from diverse fields.
- Explore cutting-edge technologies in AI Digital Afterlife & Resurrection.
- Contribute your unique insights to shape future concepts.
- Showcase your skills and engage in meaningful discussions.

The Agenda:

1. Introduction and Thesis Overview: Get a deep dive into the thesis and the latest technologies in the field.
2. Multidisciplinary Teams: Join a team of scholars.
3. Crafting Multiple Futures: Engage in a brainstorming session to envision innovative ideas.
4. Idea Selection and Development: Work with your team to refine and develop a promising concept.
5. Presentation and Critique: Showcase your team's ideas and engage in constructive discussions.

The project outlined ensures compliance with the EU's General Data Protection Regulation (GDPR) by emphasizing transparency, fairness, and lawful processing of participant data. Demographic information, discussions, and workshop recordings will be collected for necessary research purposes, with data anonymized and securely stored on the researcher's personal computer. Privacy measures, including face blurring and restricted access, will be implemented, and data-sharing agreements within the research group will be established.

Voice/video recordings aim to facilitate information extraction, with participants having the right to refuse storage or withdraw consent without providing reasons. Participation is voluntary, and discontinuation won't impact future care. The researcher, Aashay Shirsawade, is accountable for GDPR compliance, with a designated research supervisor, Sophie Sydow, email: sophie.landwehr.sydow@im.uu.se., available for complaints.

The project prioritizes GDPR adherence, participant privacy, and voluntary participation while maintaining transparency and accountability throughout the research process.

The person responsible for the project is,
Aashay Shirsawade.
Email: aashay-ajit.shirsawade.2940@student.uu.se

To show your interest, please fill out the form below.

11.3 Other team idea progressions

11.3.1 Team A

A. Abstract Entity artefacts



B. Animal artefacts





C. Gaming environment



11.3.2 Team B – AR/VR artefacts





11.3.3 Team C – Car AI

